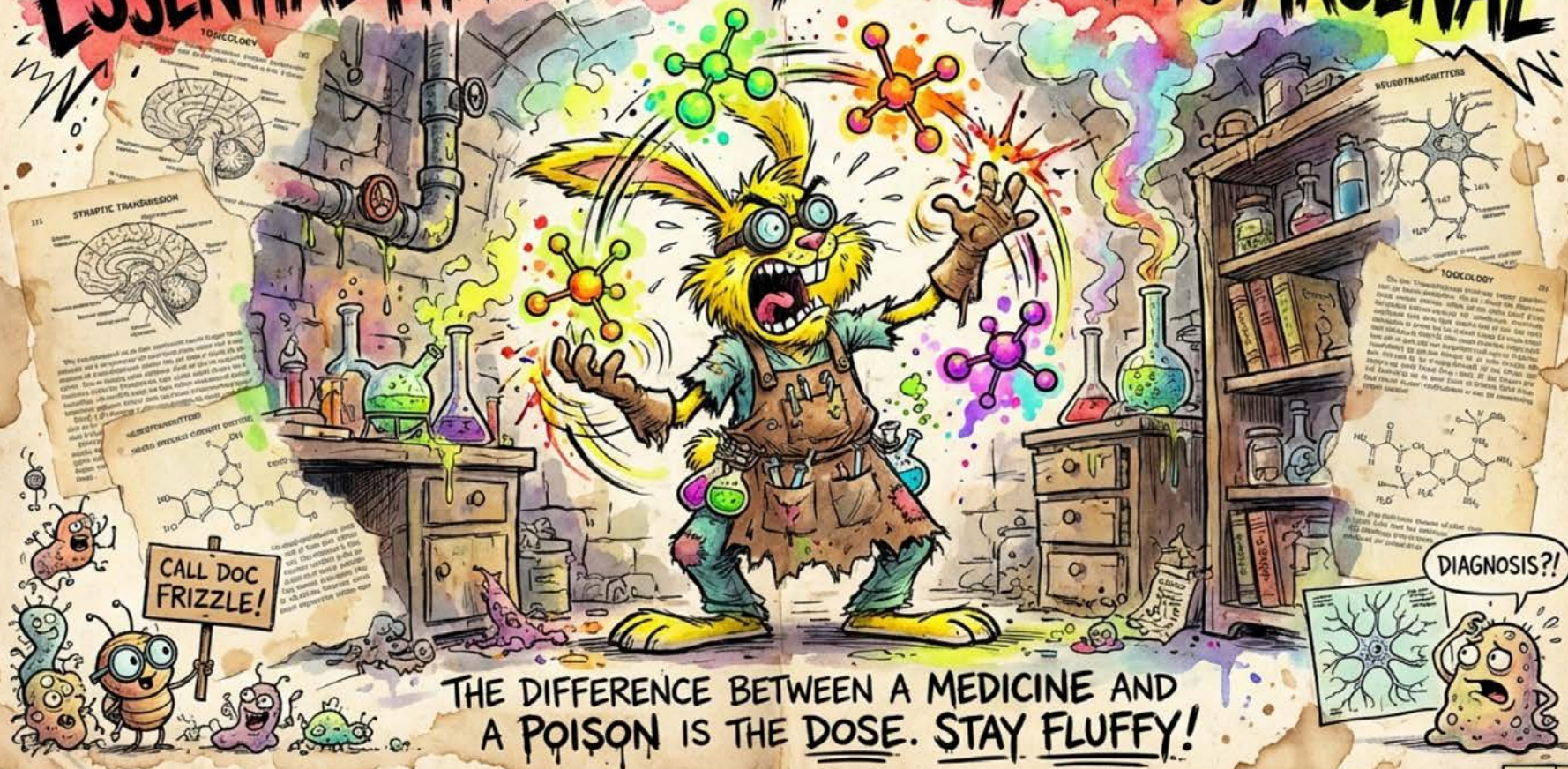


Georg von Westphalen

ESSENTIAL PHARMACOLOGY: THE SYNAPTIC ARSENAL



Lisino-the-Pril (ACE Inhibitor)

STAT: Inhibits ACE/
Renal Clearance



POWER: The Pressure Drop. LORE: Triggers Bradykinin (Cough). Watch for Angio-Edema-Annie & hyperkalemia.

Amlodi-the-Pine (Calcium Channel Blocker)

STAT: Dihydropyridine/
L-type CCB



POWER: The Pipe-Relaxer. LORE: Targeted at vessels, not the heart. Causes Edema-Eddie (swollen ankles).

Metopro-the-Lol (Beta-Blocker)

STAT: Beta-1 Selective/
Cardioselective



POWER: The Heart-Brake. LORE: Excellent for Ischemic-Ache. Masks Hypo-Harry; triggers Asthma-Arthur if selectivity fails.

Lasix-the-Loop (Furosemide)

STAT: Na-K-2Cl symport
inhibitor



POWER: The Flood-Gate. LORE: High-potency. Steals potassium (Hypo-K-Karl) and risks Oto-Toxic-Oliver at high doses.

KayfaBINGO!
Watch the
kidneys!

Aspirin-the-Ace (Antiplatelet)

STAT: Irreversible
COX-1 & 2 Inhibitor



POWER: The Slip-and-Slide. LORE: Lifetime commitment for Coronary-Canyons. Irritates Percy-the-Ulcer.

Heparin-the-Heavy (UFH/LMWH)

STAT: Activates
Antithrombin III



POWER: The Instant-Thin. LORE: UFH needs the aPTT-Check. Watch out for HIT-the-Hunter.

Rivaro-the-Ban (DOAC)

STAT: Direct Factor
Xa Inhibitor



POWER: The Factor-Freeze. LORE: No INR monitoring. Hard to reverse (needs Andexxa). Renal failure toxicity.

Warf-the-Old-Guard (Warfarin)

STAT: VKORC1 Inhibitor
(Factors 2,7,9,10)



POWER: The Slow-Bleed. LORE: Requires the INR-Tether. Vulnerable to Food-Fights (greens). Teratogenic!

HUMBUG!
Food Fights!

Metformi-the-Base (Biganide)

STAT: Inhibits Hepatic Gluconsogenesis

09



POWER: The Sugar-Stopper.

LORE: No hypoglycemia risk! Rare but lethal Lactic-Lucy in renal failure.



Insu-the-Key (Insulin)

STAT: Rapid, Intermediate, Long-acting

10



POWER: The Cell-Opener.

LORE: The Final-Boss for Type 2. Hypo-Harry is always lurking. Rotate sites!

Atorva-the-Statins (HMG-CoA Reductase Inhibitor)

STAT: Inhibits Cholesterol synthesis

11



POWER: The Gunk-Cleaner.

LORE: Prevents the Clot-Pop. Can cause Myo-Muscle-Ache and Liver-Limp.

Levothyro-the-Pulse (T4 Replacement)

STAT: Synthetic Thyroxine

12



POWER: The Metabolism-Spark.

LORE: Take on an empty stomach! Excess causes Hyper-Hanna (palpitations).

13 Morphine-the-Monarch (Opioid)

STAT: Mu-Opioid
Receptor Agonist



POWER: The Pain-Eraser.

LORE: Gold standard for agony. Causes Consti-the-Sloth & Breathing-Ben's depression.

14 Ibupro-the-Brake (NSAID)

STAT: Reversible
COX-1 & 2 Inhibitor



POWER: The Fire-Extinguisher.

LORE: Eats the stomach (Percy-the-Ulcer) and hits the kidneys (Renal-Ray).

15 Paraceta-the-Ghost (Acetaminophen)

STAT: Central COX
inhibition



POWER: The Fever-Killer.

LORE: Gentle on stomach, but has a Liver-Lethal limit (NAPQI toxicity).

16 Gabapenti-the-Numb (Neuropathic Ally)

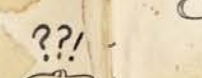
STAT: Voltage-gated
Ca²⁺ Channel ligand



POWER: The Nerve-Quietener.

LORE: Not for acute pain. Causes Dizzy-Dan and massive sleepiness.

What the
FLUFF?!



17 Amoxi-the-Standard
(Penicillin)

STAT: Beta-Lactam



POWER: The Wall-Wrecker.

LORE: Watch for Allergy-Annie and Clostrid-the-Colitis (C. diff).

19 Vancomy-the-Vault
(Glycopeptide)

STAT: Inhibits D-Ala-D-Ala
(Gram+ only)



POWER: The MRSA-Masher.

LORE: Needs Trough-Levels to save kidneys. Watch for Red-Man-Ray.

18 Ceftriaxo-the-Heavy
(3rd Gen Cephalosporin)

STAT: Great CNS penetration



POWER: The Meningitis-Mauler.

LORE: Go-To for serious Gram+/Gram- infections. Causes Biliary-Sludge.

20 Azithro-the-Macrolide

STAT: 50S Ribosome Inhibitor



POWER: The Protein-Sapper.

LORE: The Z-Pak classic. Prolongs the QT-Interval-Clock!

BLÖDSINN!

GW

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21 Salbuta-the-Rescue

STAT: Beta-2 Agonist

(SABA)



Tremor-Tim



Tachy-Tina

POWER: The Airway-Opener.

LORE: Rapid onset. Causes Tremor-Tim and Tachy-Tina.

23 Tiotro-the-LAMA

STAT: Long-acting Muscarinic Antagonist

(Anticholinergic)



Dry Tongue



Bladder

POWER: The Secretion-Dryer.

LORE: 24-hour airway opener. Causes dry mouth and urinary retention.

22 Flutica-the-Steward

STAT: Inhaled Corticosteroid

(ICS)



POWER: The Swelling-Stopper.

LORE: Chronic control. Must rinse mouth to avoid Thrush-the-Fungus.



Thrush-the-Fungus

24 Predni-the-Systemic

STAT: Glucocorticoid

(Oral Steroid)



POWER: The Total-Body-Cooler.

LORE: The Double-Edged Sword. Long-term: Buffalo-Hump-Herb & Thin-Skin-Thelma.



Humped



Fragile Skin

MedKayfabe!



GW

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25

Omepra-the-PPI

STAT: Irreversible Proton Pump Inhibitor



POWER: The Acid-Lock. LORE: Superior for Percy-the-Ulcer. Long-term drops Mg^{2+} and invites C-Diff.

27

Ondansetro-the-Stopper (Antiemetic)

STAT: $5-HT_3$ Receptor Antagonist



POWER: The Vomit-Shield. LORE: Gold standard for chemo-nausea. Causes Consti-the-Sloth and QT prolongation.

26

Metoclopra-the-Mover (Prokinetic)

STAT: D_2 Antagonist / $5-HT_4$ Agonist



POWER: The Gut-Accelerator. LORE: Crosses the BBB. Can cause The-Twitch (Extrapyramidal symptoms).

28

Lactulo-the-Breeze (Osmotic Laxative)

STAT: Non-absorbable sugar



POWER: The Water-Puller. LORE: Also a Brain-Saver; traps ammonia to help Encephalo-Ed in liver failure.



LEAKING!

AW NUTS!

CURSE YOO PIPES!

29

29

Levodopa-the-ReplacerSTAT: Dopamine Precursor
(+ Carbidopa)

POWER: The Motion-Restorer.

LORE: Carbidopa protects it peripherally. Causes On-Off-Oscillations.

Donepezi-the-Memorist
(AChE Inhibitor)STAT: Acetylcholinesterase
Inhibitor

POWER: The Signal-Booster.

LORE: Fights Alzheimer-Fog. Causes Wet-Symptoms (drooling, slow heart)

30

Pheny-the-Stabilizer
(Phenytoin)STAT: Na⁺ Channel
Blocker / Zero-order

POWER: The Fire-Extinguisher.

LORE: Stops seizure firestorms. Causes Gingival-Gus (gum overgrowth)

Sumatripta-the-Migraine-MaulerSTAT: 5-HT 1B/1D
Agonist

POWER: The Vessel-Shrinker.

LORE: Not for heart disease patients—it constricts all pipes!

KayfaBIZARRO!



33

Diazepa-the-Bouncer (Benzodiazepine)

STAT: Positive Modulator
of GABA-A



POWER: The Instant-Calm.
LORE: Dangerous with Alcohol. Highly addictive.

Stay Fluffy!

34

Sertrali-the-Mood-Lifter (SSRI)

STAT: Selective Serotonin
Reuptake Inhibitor



POWER: The Hoppy-Reservoir.
LORE: Takes 4 weeks to work. Watch out for Serotonin-Syndrome.

25

Lithiu-the-Anchor (Mood Stabilizer)

STAT: Narrow Therapeutic
Index



POWER: The Seesaw-Stabilizer.
LORE: The Lithium-Level-Check is low. Toxic to Thyroid/Renal.

36

Haloperido-the-Lock (Antipsychotic)

STAT: High-potency
D2 Antagonist



POWER: The Dino-Handcuffs.
LORE: The Old Hammer. Causes Lead-Pipe-Rigidity (NMS) and The-Lean.

37

Spirono-the-Sparing (Diuretic)

STAT: Aldosterone
Antagonist



POWER: The Potassium-Guard.

LORE: Essential for failing hearts. Can cause Man-Boob-Mike.

38

Kay-the-Supplement (Potassium Chloride)

STAT: Electrolyte
Replacement



POWER: The Spark-Plug.

LORE: Replaces K-Tokens. NEVER give IV push, or the patient dies!

39

Calciu-the-Stabilizer (Calcium Gluconate)

STAT: Membrane Stabilizer



POWER: The Heart-Shield.

LORE: Protects from Hyper-K-Karl. Doesn't lower K+, just buys time.

40

Bicarbonate-the-Buffer

STAT: Alkalinizing agent



POWER: The Acid-Neutralizer.

LORE: Used in arrests or TCA overdose. Caveat: Causes a CO₂-Spike.

KayfaBOGGLE!

41
Adrenal-the-All-Star
(Epinephrine)

STAT: Alpha-1, Beta-1,
Beta-2 Agonist



POWER: The Resurrection-Bolt.

LORE: King of Code-Blue. Can cause Engine-Overheat (ischemia).

43
Atropi-the-Speed-Up
(Anticholinergic)

STAT: Muscarinic Antagonist



POWER: The Brake-Cutter.

LORE: Cuts the Vagus-Brake. Mad as a Hatter, Dry as a Bone.

42
Noradrenal-the-Pressor
(Norepinephrine)

STAT: Strong Alpha-1
Agonist



POWER: The Pressure-Jack.

LORE: Vasopressor of Choice. Leave-Them-Dead-Norepi (necrosis risk).

44
Amioda-the-Calmer
(Antiarrhythmic)

STAT: Class III (Blocks K⁺)



POWER: The Rhythm-Lock.

LORE: Broad-spectrum. Deposits in lungs (Fibro-Fiona) and Thyroid.

Diagnosis?!

45 Digoxi-the-Fox (Cardiac Glycoside)

STAT: Na/K ATPase
Inhibitor.



POWER: The Slow-Squeeze.

LORE: Narrow window. Yellow-Vision & Reverse-Tick ECG in toxicity.

47 Allopurinolo-the-Gout-Stopper

STAT: Xanthine
Oxidase Inhibitor.



POWER: The Needle-Brakeman.

LORE: Prevention only! Triggers a riot if started during an acute attack.

46 Nitro-the-Bomb (Nitroglycerin)

STAT: Vasodilator /
NO Donor.



POWER: The Preload-Dropper.

LORE: Takes the volume load off. Do NOT use with Viagra!

48 Colchici-the-Joint-Cooler

STAT: Microtubule
Inhibitor.



POWER: The Riot-Shield.

LORE: Acute treatment. Toxicity limit marked by Diarrhea-Dan.

HUMBUG!

Cyclophospha-the-Shredder

STAT: Alkylating Agent

49



POWER: The Blueprint-Ripper. LORE: Heavy artillery. Watch for Hemorrhagic-Holly (bloody bladder)—give Mesna!

Ciclospori-the-Transplant-Guardian

STAT: Calcineurin Inhibitor

51



POWER: The Army-Sleeper. LORE: Essential for transplants. Toxic to Renal-Ray.

Methotrexa-the-Melt (Antimetabolite)

STAT: DHFR Inhibitor

50



POWER: The Folate-Thief. LORE: Kills Liver-Larry and the Bone-Marrow-Bank.

Rituxi-the-B-Cell-Marker

STAT: Anti-CD20 Monoclonal

52



POWER: The Target-Painter. LORE: Used for Lymphoma/Lupus. Can wake up Hepa-B-Ghost



KayfaBIZARRO!



The Pharma-Attending

STAT: Clinical Toxicology Master



POWER: Polypharmacy-Purger.

LORE: "Treat the patient, not the side effect! Stay fluffy!"

Quest 2: The Heart-Halt

Synthesis



LORE: Patient on Digoxi/Lasix sees "Yellow-Halos", HR 30.

QUEST: Identify Toxicity-Trap & choose Reset-Tool.

Quest 1: The Pressure Crisis

Synthesis



LORE: Patient on Lisino presents with swollen lips & K+ 6.0.

QUEST: Identify the Saboteurs & choose the next tool.

Quest 3: The Clot-Conundrum

Synthesis



LORE: Patient on Warfarin has INR 10.0 and bleeding gums.

QUEST: Call the Antidote-Squad (Vit-K & FFP).

Stay fluffy - and always check the half-life!